StateFarm Distracted Driver Detection Model

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Abstract

Approached with the task of detecting distracted drivers, we studied image-based distracted driver detection models on the StateFarm Distracted Driver Detection dataset, viewing several images from the dashboard including: safe driving, texting, phone calls, and reaching behind the passenger seat. Distracted driving is a leading cause of road accidents, and automated in-cabin monitoring could enable timely driver alerts. We trained and fine tuned models from Kaggle's competition using [key choices: augmentation, transfer learning, loss], and we evaluated with a driver-disjoint 70-30 test split. Our best model achieves an accuracy of [X%]. Denote about errors These results suggest

1. Introduction

This is the introduction.

2. Literary Analysis

To find a basis for creating a model, we first started with a literary analysis to find effective methods and shortcomings. Each participant read numberous articles and summarized the teams' results.

2.1. Distracted Driving Detection using CNNs

Four CNN models were used to training their[1] collected data using transfer learning and softmax regression: VGG-16 model, AlexNet, GoogleNet, and ResNet. The best performance came from the ResNet model, performing an accuracy of 92%, out performing the other models by up to 6%. The researchers concluded that most of the incorrect classifications from the models were due to the similarities in postures, such as drinking, hair and makeup, and texting on the phone with the left hand. One possible implementation to help reduce the number of misclassifications is to combine their overall approach with a face-based approach or to replace the final fully connected layers with classifiers such as SVM or HMM.

2.2. Distracted Driving Monitoring with Artificial Intelligence

The article[3], written by Hui Xiong and Zhiming Zhang (2026). The authors note in their overall abstract for the paper, that one of the leading causes of traffic accidents occur due to distracted driving. In the final experiment, the total F1 score used to measure performance of the model is represented by comparing the correct predictions to the actuality of the dangerous driving. In the five types of driving tested, normal, speeding, hard-breaking, distracted, and drowsy driving, all F1 scores exceed 90% with the exception of hard braking receiving an F1 score of 88.48%. The system has an overall high performance, especially with consideration to the abnormal driving behaviors. Xiong and Zhang were able to implement a system to test for distracted driving in the form of a behavioral monitoring system through the use of machine learning and computer vision techniques alongside artificial intelligence. The biggest short coming for their final model is more complex scenarios and overall road safety.

3. Training and Creating the Model

We modeled our original model from a previous submission on Kaggle created by Emmanuel Pintelas, training the model using ResNets and transfer learning. The final accuracy with three epochs was 81.9%. Between our three mememberss, we split modification of the model into error analysis & split protocol, convolution networks & saturation, and development of non-neural networks.

3.1. Split Protocol

All data was split based on the provided 26 drivers, using 70% of the set for training, and the final 30% for testing validation. The split was based on previous industry reports[1] on the best split for their training model and became the basis for ours. A random image level split allows any duplicate frames of the same driver taken from the original

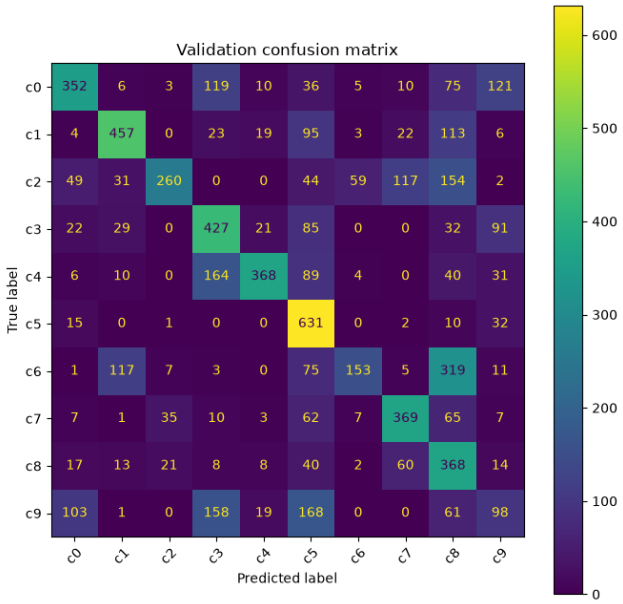


Figure 1. Original confusion matrix after training.

videos in both the training and testing data for confirmation. Using `GroupShuffleSplit` with `subject` as the group, 200 candidate seeds with the split seed DEFINE created a held-out fraction near 30% where all ten classes appear and chosen for performance. The final split uses NUMBER drivers for training (15,703 images) and NUMBER drivers for testing (6,721 images) with confirmation that the sets are disjoint.

3.2. Error Analysis

Error analysis included metrics not previously shown for the model to help better determine accuracy, bias, as well as allow us to be able to determine the best modifications for finetuning. New statistics included: entropy loss, F1 precision, and total accuracy. These were accomplished by using sklearn metrics to be computed after and during the model.

Originally, the model did not fair the best with an accuracy of 51.8%. Recall was the highest with the driver operating the radio at 0.91, and steadily declined from mid 0.6 in makeup, texing, and reaching behind range down to 0.16 when talking to the passenger. Based on how similar some of the poses are such as texting and using the phone, the recall is a significantly higher due to the clusters it creates by similarity.

The largest issue that can be denoted from this data is the inability to generalize new drivers. Not only is this denoted in the accuracy, but the high cross entropy.

4. Limitations

What we could to do improve the model.

5. Conclusion

Final conclusions.

References

[1] Yunsheng Ma, Runjia Du, Amr Abdelraouf, Kyungtae Han, Rohit Gupta, and Ziran Wang. Driver digital twin for online recognition of distracted driving behaviors. *IEEE TPAMI*, 9, 2024. Face and Gesture submission ID 324. Supplied as supplemental material `fg324.pdf`. 1

[2] Duy Tran, Ha Mahn Do, Weihua Sheng, He Bai, and Girish Chowdhary. Real-time detection of distracted driving based on deep learning. *IET*, 12, 2018.

[3] Hui Xiong and Zhiming Zhang. Design and implementation of a driving behavior monitoring system based on artificall intelligence. *IJAASR*, 5, 2026. Supplied as supplemental material `tr.pdf`. 1